



Solution Set

Question 1

Use the Maclaurin series for the exponential.

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

$$e^{2x} = 1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \dots$$

Question 2

For

$$f(x) = \sqrt{2+x}$$

around

$$x = 1$$

write

$$h = x - 1$$

so

$$\begin{aligned}\sqrt{2+x} &= \sqrt{3+h} = \sqrt{3} \left(1 + \frac{h}{3}\right)^{1/2} \\ \sqrt{2+x} &= \sqrt{3} + \frac{x-1}{2\sqrt{3}} - \frac{(x-1)^2}{24\sqrt{3}} + \frac{(x-1)^3}{144\sqrt{3}} + \dots\end{aligned}$$

Question 3

For

$$f(x) = \log(2+9x)$$

around

$$x = 1$$

write

$$h = x - 1$$

Then

$$\begin{aligned}2+9x &= 11+9h \\ \log(2+9x) &= \log 11 + \log\left(1 + \frac{9h}{11}\right) \\ &= \log 11 + \frac{9}{11}h - \frac{1}{2}\left(\frac{9}{11}\right)^2 h^2 + \frac{1}{3}\left(\frac{9}{11}\right)^3 h^3 - \dots\end{aligned}$$

Question 4

For

$$V(S) = S \log S$$
$$V'(S) = \log S + 1, \quad V''(S) = \frac{1}{S}$$

At

$$S_0 = 100, \quad \Delta S = 1$$
$$\Delta V \approx V'(100)\Delta S + \frac{1}{2}V''(100)(\Delta S)^2$$
$$\Delta V \approx (\log 100 + 1)(1) + \frac{1}{2} \frac{1}{100}(1)^2 \approx 5.61017$$